

Tracking Spatiotemporal Mineral Dissolution Through Quantitative Nuclear Magnetic Resonance Signatures

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Text S1. Reactive Transport Model: Flow, Transport and Reaction Kinetics

The reactive transport model used here is a 2D pore-scale RTM based on level-set interface tracking, rather than a purely finite-difference model. The interface evolution equation is advanced using an explicit upwind scheme on an overset Cartesian grid; flow and solute transport are solved on unstructured triangular meshes within the HyPHM framework (Frank, 2014).

The computational domain is denoted as $\Omega = [0, L_x] \times [0, L_y]$. To represent the undissolved mineral phase, we generated spatial distributions of non-overlapping circular particles with fixed radii within the domain. Calcite dissolution was simulated using physicochemical parameters for calcite, including molar volume, density, and reaction kinetics. The corresponding geometric, physical, and numerical parameters used in the simulations, including domain size, grain geometry, reaction kinetics, and time-stepping settings, are summarized in Table S1.

Within the pore fluid region, fluid motion is approximated as steady-state Stokes flow, and the governing equations are written as

$$-\nabla p + \mu \nabla^2 \mathbf{u} = \mathbf{0}. \quad (1)$$

The flow also satisfies the incompressibility constraint $\nabla \cdot \mathbf{u} = 0$. For the flow problem, a uniform injection velocity is prescribed at the inlet, a zero-stress natural boundary condition with zero reference pressure is applied at the outlet, and the lateral boundaries are treated as no-slip solid walls.

The hydrogen ion concentration field satisfies the advection–diffusion equation:

$$\frac{\partial c}{\partial t} + \nabla \cdot (\mathbf{u}c - D\nabla c) = 0. \quad (2)$$

For solute transport, the inlet is assigned a Dirichlet concentration $c = c_{\text{in}}$, the upper and lower boundaries have no-flux conditions, and the outlet is treated as a natural convective outflow. Mineral dissolution adopts a first-order linear surface reaction rate (Deng et al., 2025; Molins et al., 2021; Szawełło et al., 2024; Szymczak & Ladd, 2009):

$$R(c_{\text{H}^+}) = k_{\text{TST}} \gamma c_{\text{H}^+}, \quad (3)$$

where k_{TST} is an effective dissolution flux coefficient and γc_{H^+} is the dimensionless activity term. The interfacial reaction in its continuous form is governed by $-D\nabla c \cdot \mathbf{n} = R(c)$.

Interface retreat is described by the level-set equation:

$$\frac{\partial \phi_{\text{ls}}}{\partial t} + v_n |\nabla \phi_{\text{ls}}| = 0. \quad (4)$$

where $v_n = V_{\text{m}} R(c)$ represents the normal retreat velocity, and $\phi_{\text{ls}}(\mathbf{x}, t)$ denotes the level-set function, which distinguishes the pore space, solid regions, and transient solid–liquid interface. This equation is advanced using explicit Euler time-stepping on an overset Cartesian grid, with spatial gradients discretized via an upwind scheme. To ensure numerical stability during rapid dissolution phases, the level-set sub-timestep is constrained by the Courant–Friedrichs–Lewy (CFL) condition.

The overall numerical workflow for the pore-scale reactive transport simulation is summarized in Figure S1. Starting from an initial geometry described by the level-set function, the steady Stokes flow field is first computed. The resulting velocity field is then used to solve the advection–diffusion equation for solute transport. Based on the local concentration at the solid–liquid interface, the dissolution reaction rate and corresponding interface retreat velocity are evaluated. The level-set function is subsequently updated to

reflect the evolving geometry. This procedure is repeated sequentially until the prescribed final time is reached or complete mineral dissolution occurs.

The interaction of flow, transport, and reaction in the model is characterized by two dimensionless parameters:

$$\text{Pe} = \frac{UL}{D}. \quad (5)$$

$$\text{Da} = \frac{\gamma c_0 V_m k_{\text{TST}} L}{D}. \quad (6)$$

where γc_0 is dimensionless, L denotes the characteristic length, corresponding here to the pore-throat spacing, D is the reactant diffusion coefficient, and V_m represents the molar volume of calcium carbonate. The Péclet number (Pe) quantifies the ratio of advective to diffusive transport across the system length, whereas the Damköhler number (Da) quantifies the ratio of surface reaction to diffusion over the same scale.

Text S2. Construction of Random Grain Geometries

Random grain geometries are constructed using sequential random placement followed by iterative correction of the average pore-throat width. Candidate grain centers are sampled uniformly within the admissible domain and accepted only if they satisfy the minimum-distance constraint:

$$\|\mathbf{x}_i - \mathbf{x}_j\| \geq 2R + s_{\min}. \quad (7)$$

where $s_{\min}$ is the minimum allowable pore-throat width. The workflow enforces the target average pore-throat width $\bar{s}_{\text{tar}}$ by iteratively adjusting the target number of grains via a density factor η . After each generation round, the workflow evaluates the actual average

pore-throat width $\bar{s}_{\text{act}}$ using nearest-neighbor surface spacing and automatically increases or decreases grain density based on the deviation between $\bar{s}_{\text{act}}$ and $\bar{s}_{\text{tar}}$. When the relative error is less than 5% or the maximum number of iterations is reached, the geometry with the smallest error is retained as the final outcome.

To improve the consistency of batch simulations, the workflow saves the final random grain centers, grain radius, target average pore-throat width, and domain dimensions into a configuration file. If identical radii and similar domain sizes are used, and if the deviation between the saved and current target average pore-throat widths does not exceed a preset threshold, the workflow reuses the existing geometry rather than generating a new random one. This ensures that the same geometry is used across physical parameter cases, which minimizes the influence of random initial conditions on comparative analysis.

Once geometry generation is complete, the level-set function is constructed as the union of the grain distance fields:

$$\phi_{\text{ls}}(\mathbf{x}, 0) = \max_i (R - \|\mathbf{x} - \mathbf{x}_i\|). \quad (8)$$

Thus, $\phi_{\text{ls}} > 0$ indicates solids and $\phi_{\text{ls}} < 0$ indicates pores.

Text S3. Determination of Porosity, Permeability, and Tortuosity via Numerical Simulation

At each sampling step, the reactive transport simulation exports a set of geometric and hydraulic indices for joint analysis with subsequent NMR responses. Since the computation is two-dimensional, the code adopts an extrusion assumption with unit thickness $b = 1$, converting area and length quantities into volume and surface-area quantities, respectively.

Porosity is obtained by interpolating the current level-set function onto a regular visualization grid and calculating the fraction of pore points:

$$\varphi(t) = \frac{N_{\text{pore}}(t)}{N_{\text{tot}}}. \quad (9)$$

where $N_{\text{pore}}(t)$ denotes the number of grid points satisfying $\phi_{\text{ls}}^* \leq 0$, and N_{tot} is the total number of grid points. The initial pore volume is expressed as $PV_0 = \sum_K A_{K,\text{pore}}b$. The solid-liquid interface area is accumulated from the interface lengths within cut cells, i.e., $S(t) = \sum_K \ell_{\Gamma,K}(t)b$; the residual solid volume is obtained by subtracting the current pore area from the total domain area, then multiplying by the unit thickness.

The equivalent permeability is converted from the outlet volumetric flow rate and pressure drop via Darcy's relationship:

$$k_{\text{eff}}(t) = \frac{\mu Q(t) L_x}{A \Delta p(t)}. \quad (10)$$

where $Q(t)$ is the outlet flow rate, $\Delta p(t)$ is the average pressure difference between inlet and outlet, and $A = L_y b$ is the equivalent flow cross-sectional area. The code first outputs permeability in cm^2 , and then converts it to millidarcy (mD).

Injected pore volume is defined as the ratio of cumulative injected fluid volume to the initial pore volume, i.e., $PV_{\text{inj}}(t) = Q_{\text{in}}t/PV_0$. Here, $Q_{\text{in}} = UL_yb$ is the constant inlet flow rate. The workflow also outputs the average hydrogen ion concentration within a narrow-band region near the outlet to characterize the degree of breakthrough and reactant penetration depth.

Flow-field tortuosity is defined based on velocity ratios:

$$\tau(t) = \frac{\sum_{i \in \mathcal{P}} \|\mathbf{u}_i\|}{\sum_{i \in \mathcal{P}} u_{x,i}}. \quad (11)$$

where $\mathcal{P}$ denotes the set of velocity nodes located in the pores. This definition compares the actual velocity magnitude to the projection of velocity along the main flow direction and serves as an index characterizing the sinuosity of flow paths. In addition to these quantities, the code records the average dissolution rate and normalized permeability k/k_0 . These metrics correspond one to one with the attenuation curves and T_2 distributions derived from the NMR simulation module along a synchronized time axis, forming the dataset for the coupled analysis below.

Text S4. NMR Simulation Theory and Settings

NMR Simulation

NMR is an electromagnetic method based on the magnetic properties of atomic nuclei. Because it is sensitive to hydrogen nuclei in pore fluids, NMR uses proton-spin magnetization and relaxation in response to radio-frequency excitation at the Larmor frequency. The NMR simulation module transforms pore geometries generated by the RTM at discrete dissolution intervals into transverse magnetization decay curves and T_2 distributions. This framework follows the Bloch–Torrey formalism, which includes diffusion (Bloch, 1946; Torrey, 1956) and describes the spatiotemporal evolution of the magnetization vector $\mathbf{m} = [m_x, m_y, m_z]^T$ during excitation and relaxation:

$$\frac{\partial \mathbf{m}}{\partial t} = \gamma (\mathbf{m} \times \mathbf{B}^{\text{ex}}) - \begin{bmatrix} m_x/T_{2B} \\ m_y/T_{2B} \\ (m_z - m_0)/T_{1B} \end{bmatrix} - \nabla \cdot (D \nabla \mathbf{m}), \quad (12)$$

where γ is the gyromagnetic ratio, D represents the diffusion coefficient, and $\mathbf{B}^{\text{ex}}$ denotes the external magnetic field. At the solid–fluid interface, we impose the surface-relaxation boundary condition:

$$D \mathbf{n} \cdot \nabla \mathbf{m} + \rho_s \mathbf{m} = 0, \quad (13)$$

where $\mathbf{n}$ is the outward unit normal vector and ρ_s denotes the surface relaxivity. At the fluid–air boundary, a no-flux condition is applied:

$$D \mathbf{n} \cdot \nabla \mathbf{m} = 0. \quad (14)$$

For the NMR excitation, the static magnetic field is configured as $\mathbf{B}_0 = [0, 0, B_0]^\top$, yielding a Larmor frequency of $\omega_L = \gamma B_0$. The applied field includes a radio-frequency component and frequency offsets induced by chemical shifts and local field inhomogeneities, written as $\mathbf{B}^{\text{ex}} = [B_x, B_y, (\Delta\omega_1 + \Delta\omega_2(\mathbf{x}))/\gamma]^\top$. The resulting dynamic governing equation is cast into the following matrix form:

$$\frac{\partial}{\partial t} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} + \begin{bmatrix} 1/T_{2B} & -\Delta\omega & \omega_y \\ \Delta\omega & 1/T_{2B} & -\omega_x \\ -\omega_y & \omega_x & 1/T_{1B} \end{bmatrix} \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} - D \nabla^2 \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ m_0/T_{1B} \end{bmatrix}, \quad (15)$$

where $\Delta\omega = \Delta\omega_1 + \Delta\omega_2(\mathbf{x})$ defines the effective off-resonance frequency, and $\omega_x(t)$ and $\omega_y(t)$ represent the time-dependent pulse terms. To extract the T_2 distribution, we simulate a Carr–Purcell–Meiboom–Gill (CPMG) sequence:

$$\omega_y(t) = \begin{cases} \omega, & 0 \leq t \leq \tau \text{ or } (2n-1)\tau_1 + (2n-1)\tau \leq t \leq (2n-1)\tau_1 + (2n+1)\tau, \\ 0, & \tau < t < \tau + \tau_1 \text{ or } (2n-1)\tau_1 + (2n+1)\tau < t < (2n+1)\tau_1 + (2n+1)\tau, \end{cases} \quad (16)$$

where $n = 1, 2, 3, \dots$, ω represents the coupling strength, τ is the $\pi/2$ -pulse duration, 2τ is the ideal π -pulse duration, and τ_1 is the delay between two consecutive pulses. The measured transverse signal is then modeled as a multi-exponential decay:

$$M_{xy}(t) = M_0 \sum_i f_{2i} \exp\left(-\frac{t}{T_{2i}}\right), \quad (17)$$

where M_0 is the initial signal magnitude, f_{2i} is the fraction of the component with relaxation time T_{2i} , and the pair (T_{2i}, f_{2i}) defines the T_2 distribution.

At each temporal sampling instant, discrete attenuation curves $\{M_{xy}(t_n)\}$ are generated by integrating the transverse magnetization magnitude over the entire pore domain. The numerical solution is computed using 2D finite element discretization. To account for the evolving microstructure during dissolution, the framework extracts transient solid–fluid interfaces from reactive transport snapshots, maps these outlines to their physical dimensions, and regenerates the computational mesh. For automated simulations, boundary conditions are assigned from updated spatial coordinates rather than static geometric indices. No-flux conditions are applied to the fluid inlet and outlet, whereas active solid–fluid relaxation conditions are imposed on the lateral boundaries and internal pore interfaces. This treatment maintains boundary conformity across highly heterogeneous geometries and continuous dissolution stages. The NMR simulation and inversion settings used for these calculations are summarized in Table S2.

NMR T_2 inversion

To invert the NMR T_2 distribution, we apply a Tikhonov-regularized non-negative least squares (NNLS) algorithm (Parker & Song, 2005; Whittall et al., 1991) to the discrete attenuation curves $\{M_{xy}(t_n)\}$ obtained from the simulated magnetization relaxation signals:

$$\min_{\tilde{\mathbf{f}} \geq 0} \left\| \mathbf{W}(\tilde{\mathbf{K}}\tilde{\mathbf{f}} - \mathbf{E}) \right\|_2^2 + \varepsilon^2 \left\| \mathbf{L}\tilde{\mathbf{f}} \right\|_2^2, \quad (18)$$

where $\mathbf{W} = \text{diag}(1/\sigma_n)$ is a noise-informed weighting matrix, while $\mathbf{L}$ represents the second-difference roughness penalty. The optimal regularization parameter ε is selected

by scanning a logarithmic grid, which yields a diagnostic ratio for each candidate ε_k based on the residual energy ζ_k and structural roughness η_k :

$$R_k = \frac{\varepsilon_k^2 \eta_k}{\max(\zeta_k, \epsilon_0)}. \quad (19)$$

The optimal ε is then identified using the reciprocal-slope criterion. Further details of the inversion methodology are provided by Zou, Xie, and Arad (2016). The final outputs include the T_2 spectrum, reconstructed magnetization decay, and corresponding residuals.

These synchronized outputs allow direct comparison between geometry-derived metrics and NMR-derived relaxation features.

Text S5. Comparative Validation against Microfluidic Calcite Dissolution Experiments

To assess whether the reactive transport model reproduces the first-order geometric evolution observed in experiments, we constructed a two-dimensional simulation of the microfluidic calcite dissolution experiment reported by Rembert, Leroy, Lassin, and Roman (2024). This comparison is used as a qualitative to semiquantitative benchmark because the simulation does not resolve the full three-dimensional chip geometry, surface roughness, or all experimental heterogeneities. We therefore focus on shared observables in both systems: the direction and timing of grain retreat, porosity growth, and loss of solid–fluid interface length.

Figure S2 compares simulated snapshots with experimental images at approximately corresponding times. Both series show progressive contraction of the calcite grain, preferential retreat near the main flow path, and opening of connected pore space over a roughly 4 h interval. The simulation reproduces the dominant geometric tendency and

dissolution time scale, but it is not expected to match every local interface shape observed in the experiment.

Figure S3 compares porosity and normalized interface length or surface area. The experimental and simulated porosity curves both increase toward near-complete dissolution. Their initial values differ due to the distinct spatial domains selected for porosity calculation, with the actual mineral volumes remaining identical. After normalizing porosity growth by each initial value ($\varphi_0 = 0.71$ for the experiment and 0.80 for the simulation), dissolution progress at 1, 2, and 3 h was 0.37, 0.72, and 0.87 in the experiment and 0.36, 0.64, and 0.86 in the simulation. Across 0–4.26 h, the mean absolute difference in normalized dissolution progress was 0.028, with a root-mean-square difference of 0.037. The normalized interface metric decreases in both cases, crossing 0.5 at 2.67 h experimentally and 2.72 h numerically, and reaching zero at 4.26 and 4.28 h, respectively. The experimental perimeter shows stronger short-time fluctuations, including an early peak of $P/P_0 = 1.40$ at 0.60 h. These deviations likely reflect local roughening, imaging or segmentation sensitivity, chip-scale heterogeneity, gas-bubble effects, and thickness-dependent geometry that are smoothed or absent in the idealized two-dimensional model.

Together, Figures S2 and S3 support the model as a physically plausible two-dimensional simulation for the subsequent parameter sweeps and NMR-coupling analysis. They do not fully validate the model in a strict experimental sense; rather, they show that the model captures the dominant geometric trends required for the questions addressed here: increasing connected pore volume, decreasing reactive interface length, and dissolution

over the observed experimental time scale. The benchmark should therefore be interpreted as support for first-order rate and geometry evolution, not as a pixel-scale reproduction of the microfluidic experiment.

Text S6. T_2 -Distribution Profile Analysis

Because T_2 profiles can provide spatially resolved information for applications such as stratigraphic delineation, lithological heterogeneity identification, and groundwater-table mapping (Griffiths et al., 2024; Vang et al., 2023), we use them here to infer spatial geometric variations from NMR responses. The medium is partitioned into five flow-parallel segments, and a localized T_2 distribution is calculated for each segment. This allows us to examine spatial T_2 characteristics at permeability breakthrough across dissolution regimes.

At breakthrough, T_2 -distribution profiles show the spatial non-uniformity of dissolution (Figure S4). Channeling and wormholing show pronounced spectral amplitude at the inlet (Seg1) and lower, comparable amplitudes downstream (Seg3–Seg5), implying relatively uniform downstream water content. In contrast, face dissolution produces elevated amplitudes across Seg1–Seg4, together with peak-center shifts toward long T_2 values characteristic of free water. This indicates substantial pore-volume expansion along the flow direction. These results highlight the utility of spatially resolved T_2 distributions for evaluating dissolution heterogeneity.

Text S7. Tortuosity Profile Analysis

The segment-wise tortuosity profile for face dissolution provides the reference case for a hydraulic response controlled by front propagation (Figure 3b). In this regime, τ

first increases slightly near $\varphi \approx 0.55$ – 0.65 and reaches 1.232 at $\varphi = 0.57$, reflecting local flow deflection around the advancing dissolution front. As dissolution proceeds, the tortuosity minima and local maxima migrate progressively from Seg1 to Seg5, indicating that straightening of flow paths follows retreat of the reaction front from inlet to outlet rather than forming a conduit that spans the domain. This stepwise response supports the interpretation that the dissolved vug region and the residual matrix remain only weakly coupled until late dissolution, explaining both the persistent T_2 peak separation and the delayed permeability breakthrough in the face dissolution case.

Compared with this delayed response controlled by front propagation, channeling and wormholing show more rapid reorganization of flow paths controlled by conduits (Figure S5). The distinct initial tortuosities reflect inherent geometric heterogeneity. In the channeling regime, tortuosity in the intermediate segments (Seg2 and Seg4) declines rapidly during early dissolution, while Seg1 maintains a high initial tortuosity with minor upward fluctuations (Figure S5a). This implies that early expansion of high-velocity channels outpaces that of low-velocity regions, forcing part of the fluid to converge toward high-velocity zones through more convoluted paths. Strong convection transports sufficient reactants deep into the porous matrix, causing rapid breakthrough; subsequently, the dominant channel expands synchronously across the system, reducing tortuosity in all segments. In contrast, wormholing shows a synchronous and rapid decrease in tortuosity across all segments, suggesting that multiple highly permeable channels emerge and connect to produce rapid, relatively uniform geometric dissolution (Figure S5b). This contrast shows that tortuosity by segment separates delayed front

propagation from earlier hydraulic connection mediated by conduits, supporting the T_2 interpretations for each regime in the main text.

Text S8. Analysis of Dissolution Regimes

Our simulations span Péclet ($0.001 \leq \text{Pe} \leq 50$) and Damköhler ($0.001 \leq \text{Da} \leq 10$) numbers to map representative dissolution regimes for the selected geometry (Figure S6). Within this parameter range, varying Da alone does not produce a distinguishable transition in dissolution patterns because Da is adjusted solely by changing the inlet concentration, whereas the diffusion coefficient and surface reaction rate constant remain fixed (Wang et al., 2022). Face dissolution occurs at $\text{Pe} \leq 0.1$ and destabilizes at $\text{Pe} = 1$ and 5, eventually producing distinct fingering and multiple dominant channels at $\text{Pe} = 50$.

Text S9. T_2 Responses in Ordered Pore Networks

To test whether the simulated T_2 signatures depend on the initial pore architecture, we repeated the $\text{Pe} = 0.1$, $\text{Pe} = 1$, and $\text{Pe} = 10$ cases using an ordered grain geometry (Figure S7). This control should not be interpreted as a regime-boundary map. The emergence of channeling requires a highly heterogeneous initial pore architecture, coupled with conditions where fluid advection outpaces diffusion and the chemical reaction remains relatively slow. This advection-dominated environment allows unreacted fluids to penetrate the entire macroscopic system before being completely consumed (Leger et al., 2022). For wormholing, Wang et al. (2022) further showed that initial structural heterogeneity is an important contributor to preferential conduit development. The ordered geometry therefore suppresses visually distinct channeling and wormholing at $\text{Pe} = 1$ and $\text{Pe} = 10$, even though the transport field remains strongly Pe dependent.

Despite this morphological limitation, the results for the ordered geometry preserve the relation among advective penetration, pore coupling, permeability growth, and T_2 evolution. At $Pe = 0.1$, dissolution remains front controlled, with an approximately planar reaction front and persistent separation between the matrix pore and long- T_2 components (Figure S7a,d,g). Permeability remains limited even at $\varphi = 0.98$ ($k/k_0 = 62.2$), indicating that face dissolution is relatively insensitive to the initial pore arrangement. At $Pe = 1$, the dissolution front becomes irregular but does not form a clear channel that spans the domain; nevertheless, the long- T_2 contribution grows more rapidly, reaching 88.8% near $\varphi = 0.87$ and $k/k_0 = 94.8$ (Figure S7b,e,h). At $Pe = 10$, stronger advection distributes reactants around repeated pore throats and produces multiple branching dissolution pathways, accelerating pore coupling and permeability enhancement (Figure S7c,f,i). The breakthrough threshold is reached near $\varphi = 0.96$ ($k/k_0 = 101.7$), after which the spectra are dominated by the long- T_2 response. These results support the interpretation in the main text that increasing Pe accelerates matrix-vug coupling and T_2 peak merging, while also showing that clear channel or wormhole morphology requires sufficient initial structural heterogeneity.

Movie S1. Time evolution of the hydrogen-ion concentration field during face dissolution at $Pe = 0.1$ and $Da = 0.0369$. Time and length scales are indicated, and solid-fluid boundaries are superimposed on the concentration maps.

Movie S2. Time evolution of the hydrogen-ion concentration field during channeling at $Pe = 1$ and $Da = 0.0369$. Time and length scales are indicated, and solid-fluid boundaries are superimposed on the concentration maps.

Movie S3. Time evolution of the hydrogen-ion concentration field during wormholing at $Pe = 10$ and $Da = 0.0369$. Time and length scales are indicated, and solid-fluid boundaries are superimposed on the concentration maps.

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Table S1. Reactive transport model parameters

Category	Parameter	Symbol	Value or Description
Domain	Length in x-direction	L_x	600 μm
Domain	Length in y-direction	L_y	400 μm
Grain geometry	Grain radius	R	30 μm
Grain geometry	Designed pore-throat width	s	10 μm
Reaction	Dissolution flux coefficient	k_{TST}	$10^{-4} \text{ mol cm}^{-2} \text{ s}^{-1}$
Reaction	Activity coefficient	γ	$10^3 \text{ cm}^3 \text{ mol}^{-1}$
Reaction	Molar volume	V_{m}	$36.9 \text{ cm}^3 \text{ mol}^{-1}$
Time marching	Initial macroscopic time step	Δt_0	0.01 s
Scenarios	Control parameters	Pe, Da	Varies by case

Table S2. NMR simulation and inversion parameters

Category	Parameter	Symbol	Value or Description
Field	Magnetic field offset	B	10^{-6} T
Field	Gyromagnetic ratio	γ_{H}	45 MHz T $^{-1}$
Pulse	Primary pulse duration	τ	3.4917 μ s
Pulse	RF amplitude	B_1	0.01 T
Field	Frequency offset	ω	$\gamma_{\text{H}}B$
Relaxation	Spin-lattice time	$T_{1\text{B}}$	3000 ms
Relaxation	Spin-spin time	$T_{2\text{B}}$	3000 ms
Diffusion	Bulk diffusivity	D	2×10^{-9} m 2 s $^{-1}$
Relaxation	Surface relaxivity	ρ_s	50 μ m s $^{-1}$
Sequence	Auxiliary frequency	k	10 Hz
Magnetization	Equilibrium density	m_0	10^{12} A m $^{-3}$
Pulse	RF coupling	ω_1	$\gamma_{\text{H}}B_1$
Sequence	Echo-spacing time	T	$3.4907 \times 10^{-6} + 2 \times 0.05$ s
Sequence	Pulse reciprocal factor	a_d	$(1/3.4907) \times 10^6$ s $^{-1}$
Pulse	Nominal π coupling	ω_2	$2\gamma_{\text{H}}B_1$
Pulse	Secondary RF amplitude	B_2	10^{-5} T
Pulse	Secondary pulse duration	τ_2	3490.7 μ s
Pulse	Secondary RF coupling	ω_3	$\gamma_{\text{H}}B_2$
Inversion	Regularization parameter	ε	Logarithmic candidate scan

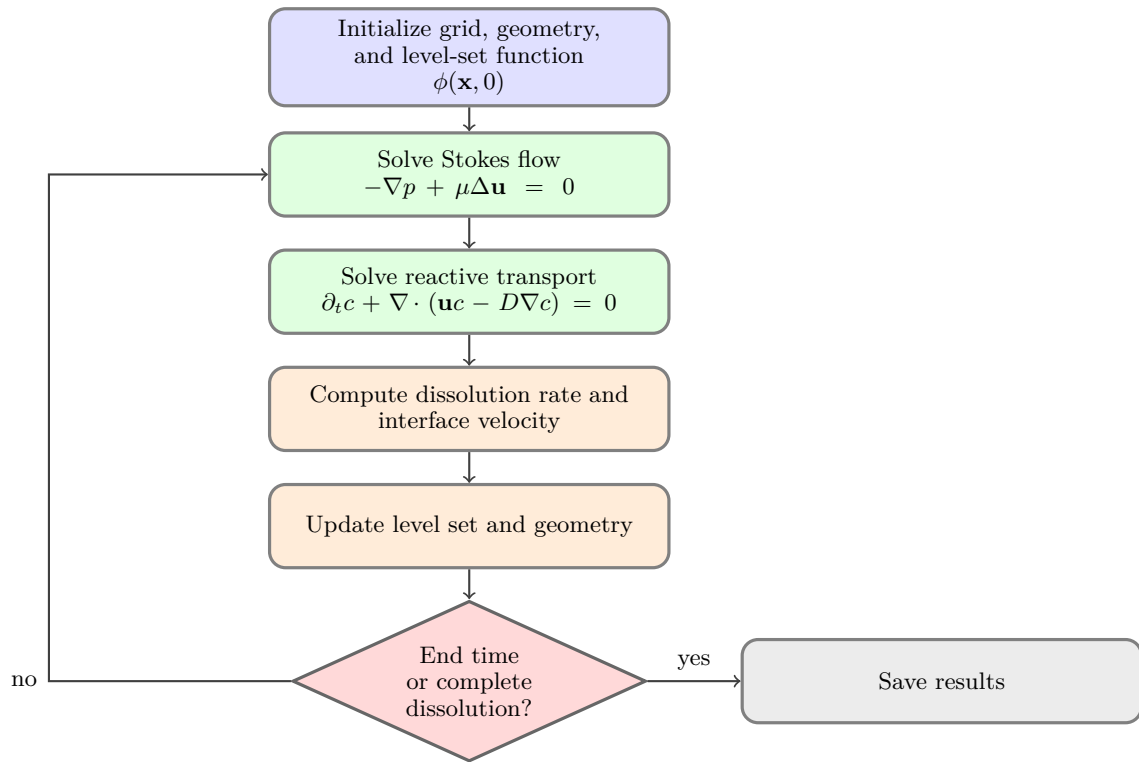


Figure S1. Pore-scale reactive transport workflow with evolving geometry. Colors denote initialization (blue), flow and transport solution (green), reaction and geometry update (orange), and output or termination (gray/red).

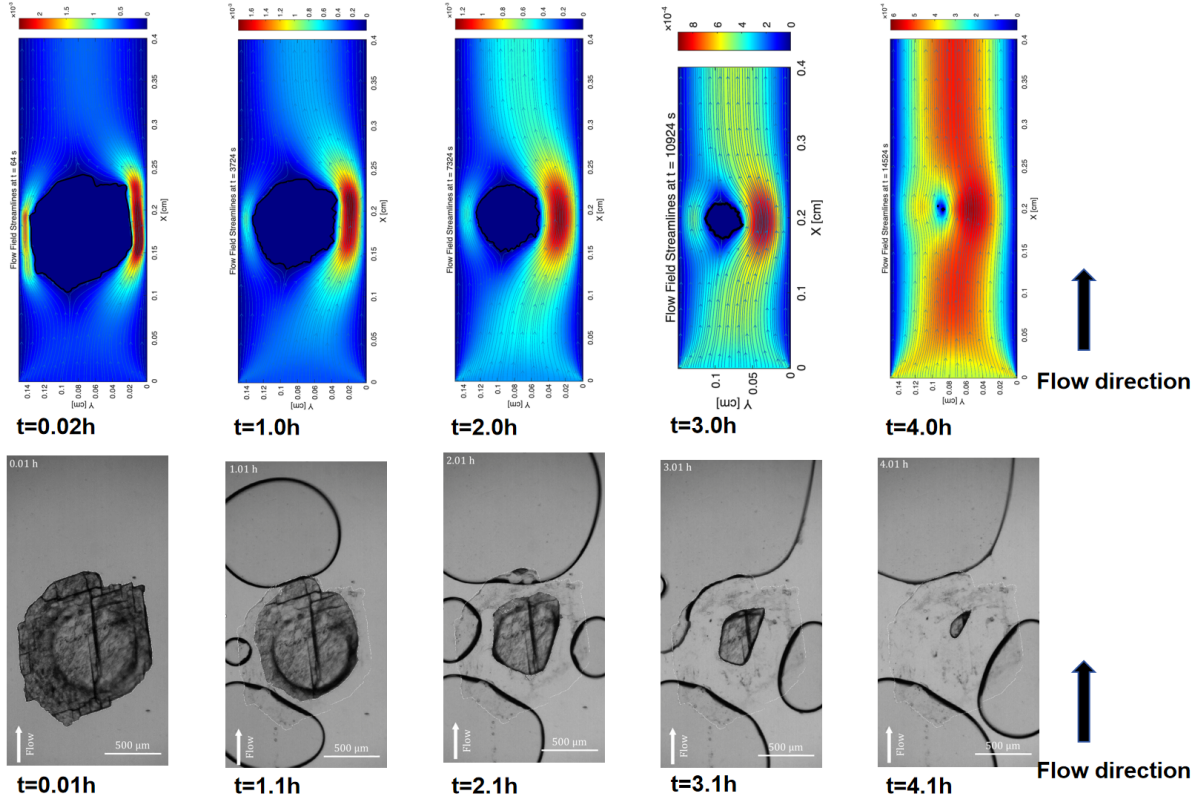


Figure S2. Morphological comparison between the two-dimensional simulation and the microfluidic calcite dissolution experiment of Rembert et al. (2024). Simulated flow fields and solid boundaries (top) and experimental images (bottom) show grain retreat along the main flow path, pore-space opening, and near-complete dissolution over about 4 h.

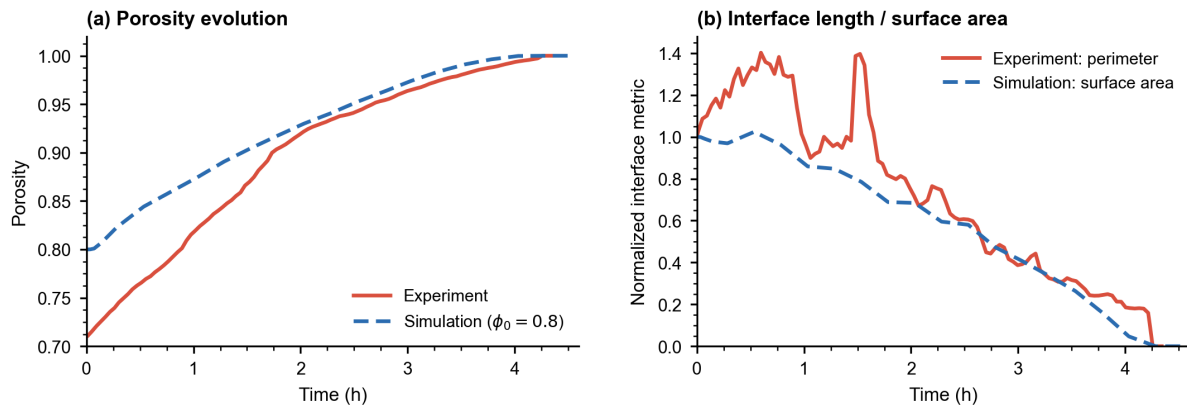


Figure S3. Geometry-derived metrics from the microfluidic experiment and simplified two-dimensional simulation. Porosity increases (left), whereas normalized interface length or surface area decreases (right). Solid and dashed curves denote experiment and simulation; agreement is interpreted at the trend level because the simulation smooths local fluctuations and omits the full three-dimensional chip.

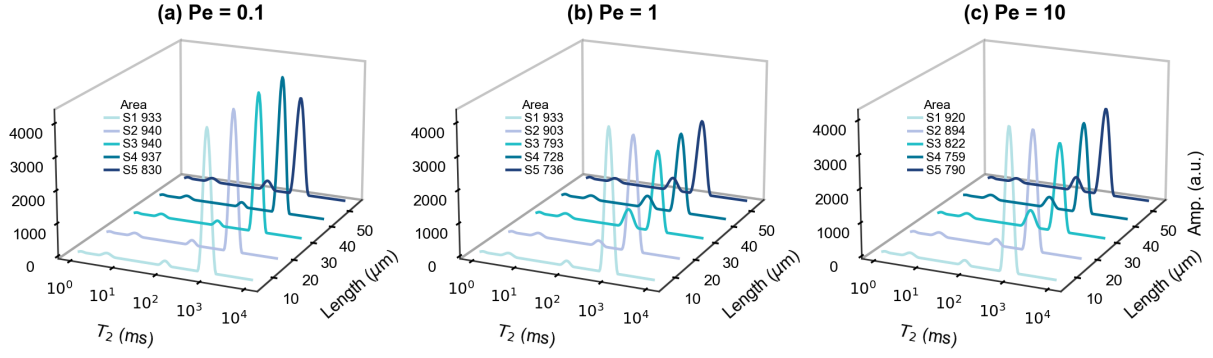


Figure S4. T_2 -distribution profiles at permeability breakthrough. Five inlet-to-outlet segmental spectra (Seg1–Seg5) are shown as waterfall plots at BT ($k/k_0 \approx 100$). **(a)** $Pe = 0.1$ (face dissolution), **(b)** $Pe = 1$ (channeling), and **(c)** $Pe = 10$ (wormholing). Legend areas are integrated signal amplitudes used as proxies for local water content or pore volume.

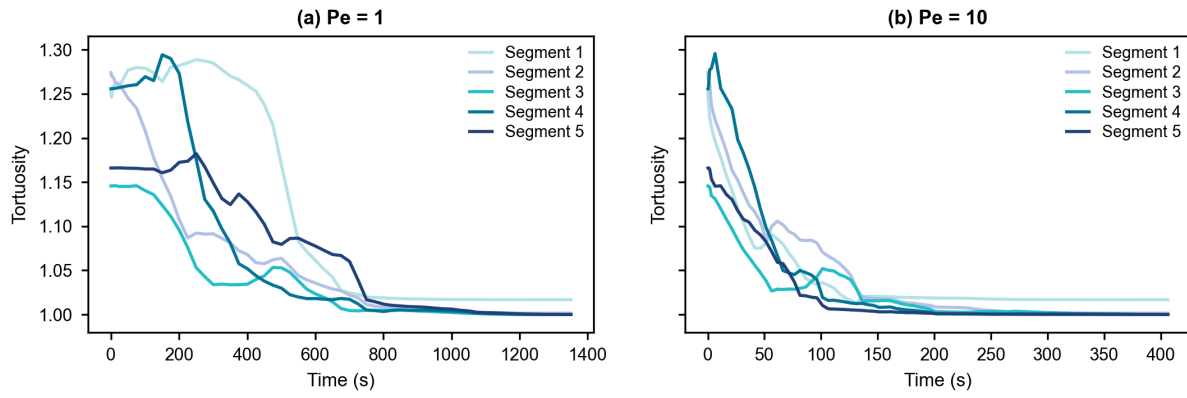


Figure S5. Segment-wise tortuosity evolution for (a) $Pe = 1$ and (b) $Pe = 10$. The domain is divided into five inlet-to-outlet segments (Seg1–Seg5), and tortuosity is computed from the instantaneous velocity field (Text S3). Near-synchronous decreases indicate global flow-path reorganization.

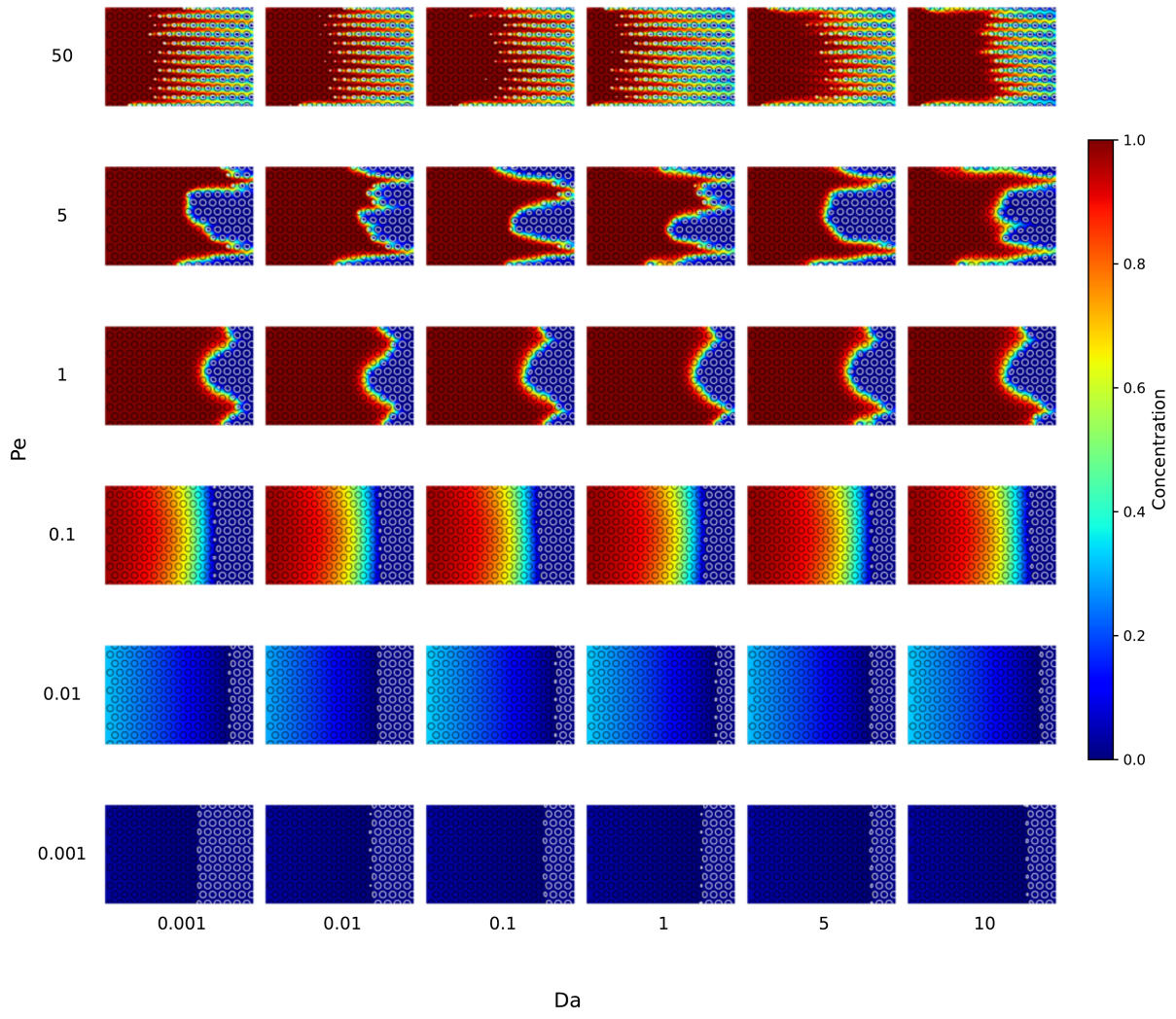


Figure S6. Dissolution patterns across the Péclet–Damköhler parameter space.

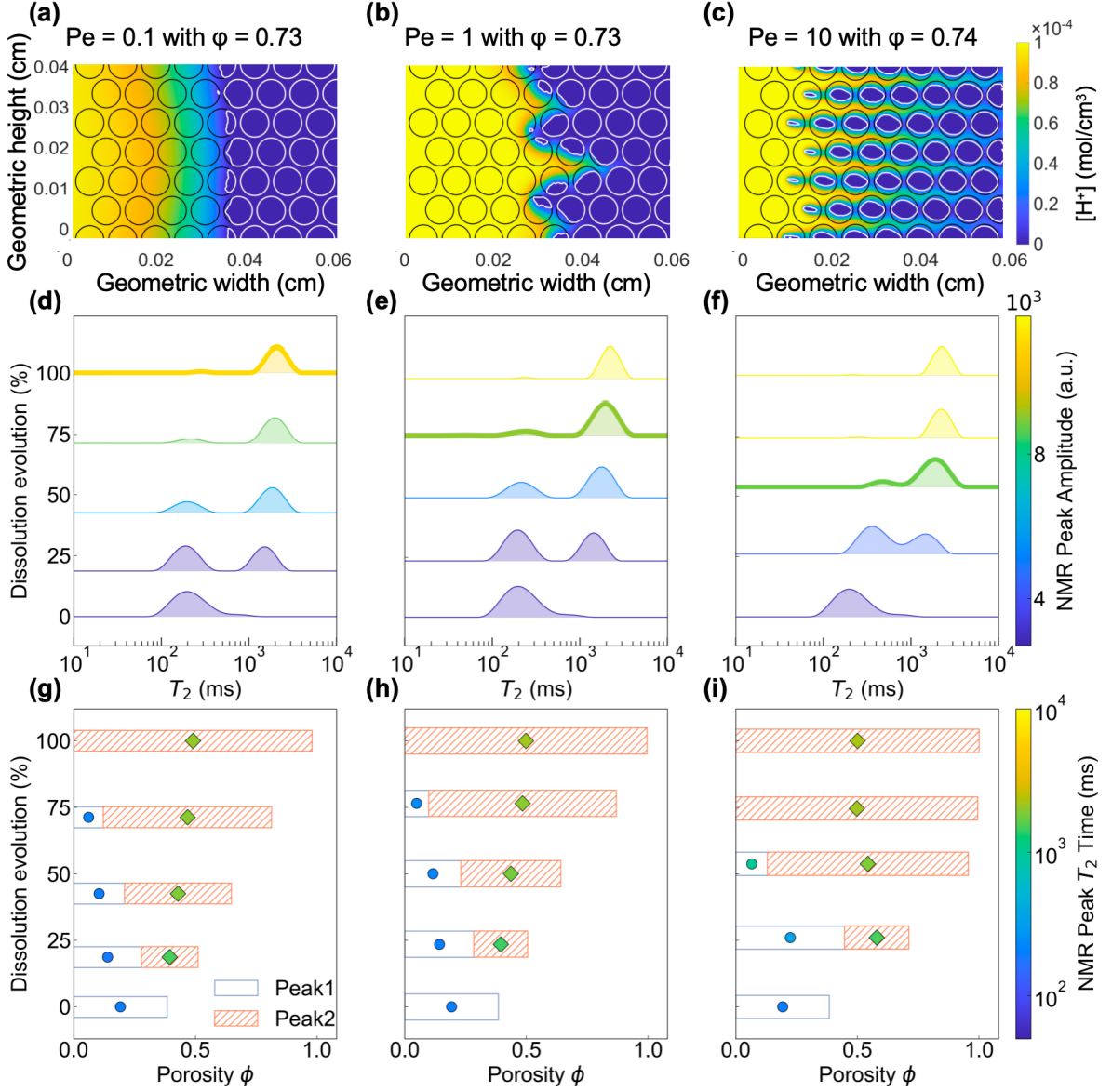


Figure S7. Ordered-geometry control simulations for evaluating structural sensitivity of the T_2 response. **(a–c)** Hydrogen-ion concentration fields at comparable porosities for $Pe = 0.1$, $Pe = 1$, and $Pe = 10$. **(d–f)** Bulk T_2 spectra during normalized dissolution evolution, with bold curves marking the time steps closest to permeability breakthrough. **(g–i)** Porosity-weighted peak-class contributions and merged peak centers, where Peak1 represents the matrix-pore response and Peak2 represents the long- T_2 dissolution-induced vug response. The ordered structure suppresses visually distinct channeling and wormholing at $Pe = 1$ and $Pe = 10$, but still shows Pe -dependent acceleration of pore coupling, permeability growth, and T_2 peak merging.